the epochalypse catalog

G23H catalog
DR2–DR3 cross-match
$G < 16$
18,061,961 stars

→ 5,789,521 stars
$\varpi > 2$ mas
($d < 500$ pc)

→ 5,729,075 stars
−60,446:
duplicated DR2–DR3
cross-match rows

→ 5,727,981 stars
−1,080: no $\sigma_{\rm AL}$ (DR3)
−14: no $\sigma_{\rm att}$ (DR3)

→ 5,040,328 stars
Pecaut & Mamajek (2013) cuts:
−3,409: $M_G \notin [-1.19, 17.3]$
−684,244: $|\Delta M_G| > 1$ or $G_{\rm BP} - G_{\rm RP} \notin [-0.12, 5.1]$

substellar companion injection
$M_p \in [1\,M_{\rm Mars},\ 80\,M_{\rm Jup}]$ log-uniform $a \in [0.03, 100]$ au log-uniform $e \sim \mathcal{U}(0, 0.99)$ isotropic angles

zero-companions
n = 5,040,328

companion-free control
the matched null

one-companion
n = 5,040,328

occurrence rate agnostic
no detectability filter

two-companions
n = 5,040,328

occurrence rate agnostic
16,615 near-resonant (2:1 or 3:2)
50/50 coplanar; $a_1 < a_2$

↑ the full epochalypse catalog

↓ high-SNR sub-populations

high-SNR
n = 436,883

$\mathrm{SNR}_{\rm tot} \geq 5$

high-SNR
n = 25,946

both companions $\mathrm{SNR}_{\rm tot} \geq 5$
272 near-resonant (2:1 or 3:2)